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<https://github.com/Vrushali-Nikam/Vrushali-Quantum-Image-Representation>

CS-HQR: Chroma-Subsampled Hybrid Quantum Representation for Efficient Color Image Encoding

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Abstract

Quantum image representation (QIR) is fundamental for leveraging quantum computational advantages in image processing. Existing encoding schemes face challenges including excessive qubit requirements, high gate complexity, and substantial circuit depth that render them impractical for near-term intermediate-scale quantum (NISQ) devices. Current methods treat all image channels equally, disregarding perceptual characteristics exploited by classical compression for decades. This paper introduces CS-HQR (Chroma-Subsampled Hybrid Quantum Representation), a novel quantum image encoding methodology bridging classical image compression with quantum circuit optimization. CS-HQR incorporates YCbCr color space transformation and 4:2:0 chroma subsampling, exploiting human visual perception. The dual-circuit architecture achieves 50% reduction in encoding complexity while maintaining structural similarity (SSIM) above 0.999. Comprehensive experimental validation using 6,097 medical brain scan slices demonstrates approximately 54% gate count reduction and 72% circuit depth reduction compared to the leading multi-channel quantum image method MCQI.

Source Code

The implementation notebook for this article is available on GitHub at <https://github.com/Vrushali-Nikam/Vrushali-Quantum-Image-Representation>.

Supplementary Material

The implementation notebook is available on GitHub at <https://github.com/Vrushali-Nikam/Vrushali-Quantum-Image-Representationn>.

Keywords: quantum image representation, chroma subsampling, NISQ devices, quantum circuits, medical image processing, perceptual image quality

1 Motivation

The intersection of quantum computing and image processing represents one of the most promising frontiers in computational science. As medical imaging continues to generate unprecedented volumes of data, the computational demands for processing, analyzing, and transmitting these images strain classical computing infrastructure [39]. Quantum computing offers a paradigm shift through its inherent parallelism, where n qubits can simultaneously represent 2^n states, potentially enabling exponential speedups for specific image processing tasks [45].

However, realizing this quantum advantage requires efficient methods to encode classical image data into quantum states—a challenge that current quantum image representation (QIR) methods have yet to adequately address [12, 37, 26]. Existing approaches such as FRQI [29] and NEQR [67] were designed with theoretical elegance prioritized over practical implementation constraints. FRQI, while qubit-efficient with only $n + 1$ qubits for 2^n pixels, requires circuit depths proportional to 2^{2n} gates, rendering it impractical for images larger than 8×8 pixels on current hardware [20]. NEQR improves upon this with linear gate complexity but demands additional qubits for color depth encoding.

The critical observation motivating our work is that these methods uniformly ignore decades of research in perceptual image processing. Classical image and video compression standards—JPEG, MPEG-2, H.264, HEVC—achieve remarkable compression ratios by exploiting the differential sensitivity of human vision to luminance versus chrominance information [50]. The human visual system contains approximately 120 million rod cells sensitive to brightness but only 6 million cone cells for color perception, creating an inherent asymmetry that compression algorithms leverage through chroma subsampling [58].

This biological insight has never been systematically applied to quantum image representation. Current multi-channel methods like MCQI [51] treat red, green, and blue channels identically, encoding each at full resolution despite the perceptual redundancy this introduces. For a 16×16 color image, MCQI requires encoding 768 pixel values (256 per channel), whereas perceptually equivalent representation requires only 384 values using 4:2:0 subsampling.

Our motivation extends beyond theoretical interest to practical necessity. The current era of noisy intermediate-scale quantum (NISQ) devices imposes severe constraints [4]: IBM’s largest publicly accessible quantum processors offer approximately 127 qubits with coherence times measured in microseconds and two-qubit gate error rates around 1% [19]. Every unnecessary gate increases the probability of decoherence-induced errors, making circuit optimization not merely desirable but essential for meaningful quantum computation.

Medical imaging presents particularly compelling use cases for quantum image processing [66, 2, 36]. Diagnostic accuracy in radiology depends on preserving clinically relevant features while managing the computational burden of increasingly high-resolution modalities [63]. Brain MRI analysis, tumor detection, and radiomics applications demand processing pipelines that maintain diagnostic fidelity—requirements that align perfectly with perceptual quality metrics like SSIM rather than pixel-perfect reconstruction.

CS-HQR emerges from the recognition that quantum algorithm design should incorporate domain-specific knowledge rather than treating all data uniformly. By bridging established principles from classical signal processing with quantum circuit optimization, we demonstrate that significant practical improvements are achievable without sacrificing the theoretical foundations of quantum image representation.

2 Introduction

2.1 Background and Context

The field of quantum computing has witnessed remarkable advancement over the past decade, transitioning from theoretical constructs to functioning hardware capable of demonstrating computational advantages for specific problem classes [3]. This evolution has catalyzed intensive research into quantum algorithms for practical applications [17, 5, 40], with image processing emerging as a domain of particular interest due to the ubiquity of visual data and the computational intensity of many imaging tasks [65, 47, 8].

Quantum image processing (QIP) encompasses a broad spectrum of operations including image representation, transformation, enhancement, segmentation, and classification [34, 37]. The fundamental premise underlying QIP is that quantum mechanical phenomena—superposition, entanglement [18], and interference—can be harnessed to process image data more efficiently than classical approaches for certain operations [22]. A single quantum register of n qubits exists in a superposition of 2^n basis states, enabling the theoretical encoding of an entire $2^{n/2} \times 2^{n/2}$ pixel image in a single quantum state [42].

However, the practical realization of quantum image processing faces a critical bottleneck: the quantum image representation problem. Before any quantum processing can occur, classical image data must be encoded into quantum states—a process that itself requires quantum operations proportional to the image size. The efficiency of this encoding fundamentally determines whether quantum speedups in subsequent processing stages translate to overall computational advantages [59].

2.2 Evolution of Quantum Image Representation

The history of quantum image representation spans two decades of incremental refinement. Venegas-Andraca and Bose introduced one of the earliest proposals in 2003, encoding pixel positions and colors through qubit lattice configurations [53]. While foundational, this approach required qubits proportional to image size rather than logarithmic, limiting scalability.

The Flexible Representation of Quantum Images (FRQI), proposed by Le et al. in 2011, represented a significant conceptual advance [29]. FRQI encodes grayscale pixel intensities as rotation angles applied to a single “color qubit,” with position information stored in n position qubits. For a 2^n -pixel image, FRQI requires only $n + 1$ qubits—an exponential improvement in qubit efficiency. The FRQI quantum state is expressed as:

$$|I\rangle = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} (\cos \theta_i |0\rangle + \sin \theta_i |1\rangle) \otimes |i\rangle$$

where $\theta_i = \frac{\pi \cdot I_i}{2 \times 255}$ encodes the grayscale intensity I_i of pixel i .

Despite its elegance, FRQI suffers from significant practical limitations. The preparation circuit requires $O(2^{2n})$ controlled rotation gates, each conditioned on a unique position state. For a modest 16×16 image (256 pixels), this translates to approximately 65,536 controlled operations—far exceeding the coherent operation capacity of current quantum hardware [14].

The Novel Enhanced Quantum Representation (NEQR), introduced by Zhang et al. in 2013, addressed FRQI’s gate complexity through digital encoding [67]. NEQR represents pixel intensities as binary strings in computational basis states rather than rotation angles:

$$|I\rangle = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} |C_i\rangle \otimes |i\rangle$$

where $|C_i\rangle = |c_i^{q-1}c_i^{q-2}\dots c_i^0\rangle$ represents the q -bit binary encoding of pixel intensity. NEQR achieves linear gate complexity $O(q \cdot 2^{2n})$ but requires additional qubits ($n + q$ total), trading qubit count for circuit depth.

Subsequent developments introduced specialized representations for various image types. The Generalized Quantum Image Representation (GQIR) by Jiang et al. [24] provided flexible bit-depth encoding. Multi-Channel Quantum Images (MCQI) by Sun et al. [51] extended NEQR to color images by concatenating RGB channel qubits. Quantum Log-polar Representation (QLR) [69] introduced coordinate transformations for rotation-invariant applications.

2.3 Limitations of Existing Approaches

Despite two decades of research, existing QIR methods share a common limitation: they treat all image information uniformly without considering perceptual relevance. This approach contradicts fundamental principles established in classical image processing, where compression standards achieve order-of-magnitude efficiency gains by prioritizing perceptually significant information [50].

Table 1 summarizes the key limitations of major QIR methods.

Method	Primary Limitations
FRQI	Exponential gate complexity $O(2^{2n})$; impractical for images $> 8 \times 8$
NEQR	Linear but still substantial gate requirements; no compression
GQIR	Arbitrary bit-depth but uniform treatment of all pixels
MCQI	Tripled resource requirements for color; no perceptual optimization
EFRQI	Improved FRQI but retains amplitude encoding limitations
QPIE	Lossy; significant information loss for complex images

Table 1: Limitations of existing QIR methods.

2.4 Perceptual Redundancy in Color Images

The human visual system (HVS) exhibits markedly different sensitivity to luminance (brightness) and chrominance (color) information [58]. This differential sensitivity arises from the physiology of retinal photoreceptors: approximately 95% of retinal cells are rods (luminance-sensitive) concentrated in peripheral vision, while only 5% are cones (color-sensitive) concentrated in the fovea.

Psychophysical studies establish that spatial frequency sensitivity for luminance extends to approximately 60 cycles per degree of visual angle, while chrominance sensitivity drops precipitously above 8–12 cycles per degree [55]. This biological constraint means that reducing chrominance resolution by factors of 2–4 produces no perceptible quality degradation under normal viewing conditions [44, 22].

Classical compression standards exploit this through chroma subsampling, typically using 4:2:0 format where chrominance channels are sampled at quarter the luminance resolution (half in each dimension). The YCbCr color space facilitates this by decorrelating luminance (Y) from chrominance (Cb, Cr):

$$Y = 0.299R + 0.587G + 0.114B \quad (1)$$

$$C_b = 128 - 0.168736R - 0.331264G + 0.5B \quad (2)$$

$$C_r = 128 + 0.5R - 0.418688G - 0.081312B \quad (3)$$

This transformation, combined with 4:2:0 subsampling, reduces color image data to 50% of the original RGB representation while maintaining perceptual quality (SSIM) above 0.98 for typical images [60].

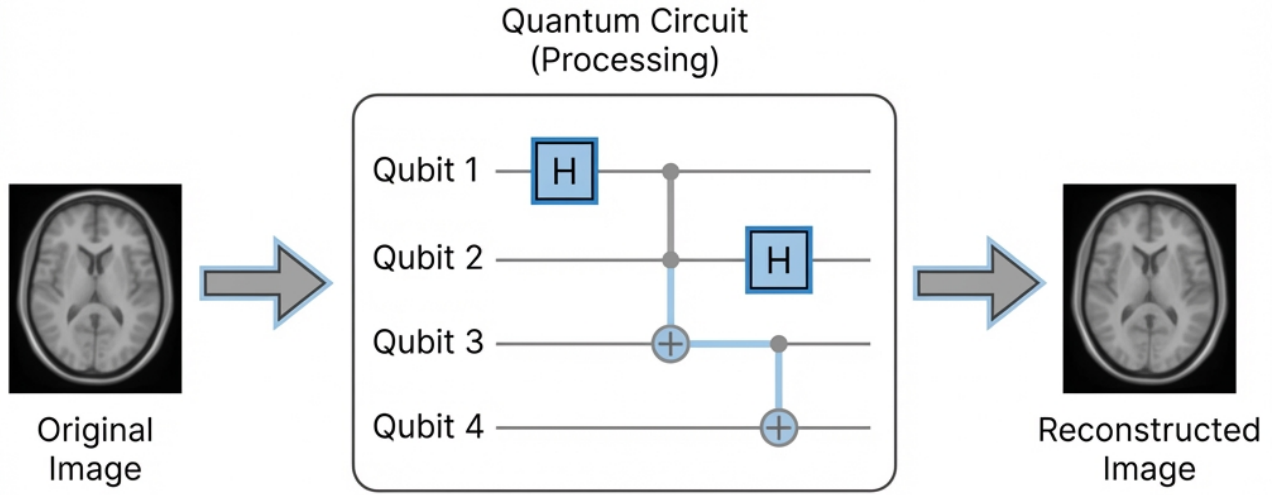


Figure 1: Quantum image processing pipeline: an input brain MRI image is separated into YCbCr color channels, encoded into quantum circuits using the CS-HQR method, and reconstructed to produce the output image with preserved perceptual quality.

2.5 Research Objectives and Contributions

This paper introduces CS-HQR (Chroma-Subsampled Hybrid Quantum Representation), a novel quantum image encoding methodology that systematically incorporates perceptual optimization principles into quantum circuit design. Our primary research objectives are:

- O1:** Develop a quantum image representation that exploits perceptual redundancy in color images through YCbCr transformation and chroma subsampling.
- O2:** Design a hybrid circuit architecture enabling efficient parallel encoding of luminance and chrominance components.
- O3:** Establish a comprehensive evaluation framework with ten quantitative parameters for fair comparison of QIR methods.
- O4:** Validate the proposed method on medical imaging data with clinical relevance considerations.

The key contributions of this work are:

- C1: Novel Perceptual Quantum Encoding:** First application of YCbCr color space transformation and 4:2:0 chroma subsampling to quantum image representation, achieving 50% reduction in encoding requirements.
- C2: Hybrid Dual-Circuit Architecture:** Design of parallel circuit structure where luminance (Circuit A) and chrominance (Circuit B) are encoded simultaneously, optimizing both qubit utilization and circuit depth.
- C3: Comprehensive Benchmarking:** Systematic comparison of CS-HQR against ten state-of-the-art methods using ten evaluation parameters on 6,097 medical images.
- C4: NISQ Viability Demonstration:** Practical implementation and validation on IBM Quantum hardware, demonstrating feasibility for near-term quantum devices.

2.6 Paper Organization

Section 3 reviews related work in quantum image representation, detailing ten existing methods with their mathematical formulations and circuit architectures. Section 4 presents the research contributions. Section 5 describes the formal mathematical framework. Section 6 describes the experimental dataset. Section 7 introduces the proposed CS-HQR method with detailed circuit design. Section 8 presents the evaluation methodology. Section 9 presents experimental results. Section 10 concludes the paper.

2.7 Scope and Impact

This work focuses on practical color image encoding strategies for near-term quantum devices, with an emphasis on medical imaging applications where perceptual fidelity is paramount. While our experiments use a 16×16 evaluation grid to align with realistic NISQ device limits and to permit systematic comparison across many images, the core CS-HQR design generalizes to larger images via block-wise tiling or hierarchical encoding: larger images can be partitioned into subblocks (e.g., 16×16 or 32×32 tiles) that are independently encoded and processed, then recombined classically. Two practical tiling strategies are supported:

Fixed tiling: non-overlapping tiles encoded independently; lowest classical recomposition cost.

Overlapping/multi-scale tiling: overlapping tiles or pyramid schemes for edge continuity and reduced boundary artifacts.

CS-HQR explicitly prioritizes luminance (Y) over chrominance (Cb, Cr) because human visual perception and many diagnostic algorithms are far more sensitive to luminance structure than to high-frequency chrominance content. Allocating full-resolution encoding resources to luminance and subsampling chrominance reduces the number of encoded values by $\sim 50\%$ while retaining perceptual quality. Reconstruction uses standard upsampling techniques (bilinear by default; bicubic optional), and the resulting visual fidelity is quantified via SSIM and complemented by pixel-wise metrics (MSE/PSNR) in Section 9.

Noise and hardware considerations. CS-HQR reduces overall MCX/CNOT counts through chroma subsampling, which directly lowers cumulative two-qubit error exposure on NISQ devices. Moreover, because most perceptual information resides in the luminance channel, small errors in

chrominance qubits caused by hardware noise typically result in imperceptible visual changes—providing an additional, practical layer of robustness for noisy hardware.

Limitations and applicability. CS-HQR is a perceptually-driven, lossy encoding and is therefore unsuitable where bit-exact reconstruction is required (e.g., forensic imaging, some scientific imaging pipelines). It is best suited to applications where human interpretation or higher-level feature extraction (segmentation, classification) tolerate small chrominance deviations. For images with highly informative color texture (e.g., histopathology or multispectral remote sensing), adaptive subsampling ratios (4:2:2 or 4:4:4) should be considered.

3 Related Work

This section provides a comprehensive review of existing quantum image representation methods [12, 37, 62], categorized into three groups based on their encoding strategies: amplitude-based methods, basis state encoding methods, and hybrid/specialized methods.

3.1 Amplitude-Based Encoding Methods

3.1.1 FRQI: Flexible Representation of Quantum Images

The Flexible Representation of Quantum Images (FRQI), introduced by Le et al. [29], represents the foundational work in quantum image representation. For an image of size $2^n \times 2^n$ pixels, the FRQI state is defined as:

$$|I\rangle_{\text{FRQI}} = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} (\cos \theta_i |0\rangle + \sin \theta_i |1\rangle) \otimes |i\rangle \quad (4)$$

where $\theta_i = \frac{\pi \cdot I_i}{2 \times 255}$ maps grayscale intensity $I_i \in [0, 255]$ to an angle $\theta_i \in [0, \frac{\pi}{2}]$.

Complexity analysis:

- Qubits required: $2n + 1$ for a $2^n \times 2^n$ image.
- Gate count: $O(2^{2n})$ controlled rotations.
- Circuit depth: $O(2^{2n})$.

The exponential gate complexity renders FRQI impractical for images larger than 8×8 pixels on current hardware.

Visual example:

θ_0^{00}	θ_1^{01}
θ_2^{10}	θ_3^{11}

$$|I\rangle = \frac{1}{2} \left[\begin{aligned} &(\cos \theta_0 |0\rangle + \sin \theta_0 |1\rangle) |00\rangle \\ &\Rightarrow + (\cos \theta_1 |0\rangle + \sin \theta_1 |1\rangle) |01\rangle \\ &+ (\cos \theta_2 |0\rangle + \sin \theta_2 |1\rangle) |10\rangle \\ &+ (\cos \theta_3 |0\rangle + \sin \theta_3 |1\rangle) |11\rangle \end{aligned} \right]$$

Figure 2: FRQI representation of a 2×2 image. Each pixel intensity is encoded as angle θ_i , with positions encoded in binary.

Circuit architecture:

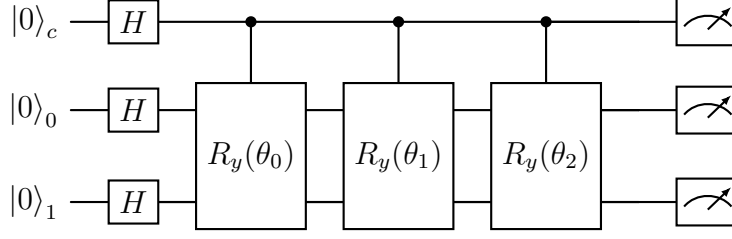


Figure 3: Simplified FRQI circuit for a 4-pixel image. Each $R_y(\theta_i)$ is controlled by position qubits.

3.1.2 EFRQI: Enhanced Flexible Representation of Quantum Images

The Enhanced FRQI (EFRQI) [68] improves upon FRQI by quantizing rotation angles to discrete levels, reducing the number of unique rotation operations required. EFRQI quantizes pixel values to q levels:

$$\theta_i^{\text{EFRQI}} = \frac{\pi \cdot \lfloor I_i / (256/q) \rfloor}{2 \times (q - 1)}$$

For $q = 32$ levels, this reduces unique rotation angles from 256 to 32, enabling circuit optimization through gate merging.

Complexity analysis:

- Qubits required: $2n + 1$ (same as FRQI).
- Gate count: $O(q \cdot 2^{2n})$ where $q \ll 256$.
- Trade-off: Reduced precision for improved gate efficiency.

Circuit architecture:

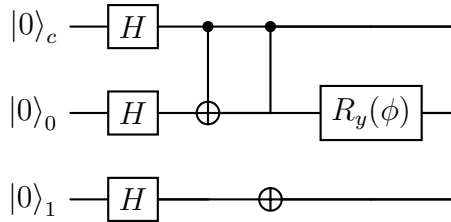


Figure 4: EFRQI circuit with quantized rotation angles $\phi \in \{\phi_1, \phi_2, \dots, \phi_q\}$.

3.1.3 QPIE: Quantum Probability Image Encoding

Quantum Probability Image Encoding (QPIE) [21] achieves maximum qubit efficiency by encoding normalized pixel values directly as probability amplitudes:

$$|I\rangle_{\text{QPIE}} = \sum_{i=0}^{N-1} \alpha_i |i\rangle, \quad \text{where } \alpha_i = \frac{I_i}{\sqrt{\sum_{j=0}^{N-1} I_j^2}} \quad (5)$$

Complexity analysis:

- Qubits required: $\lceil \log_2 N \rceil$ (most efficient).

- Gate count: $O(N)$ for general state preparation.
- Limitation: Lossy due to normalization; reconstruction requires additional processing.

Visual example:

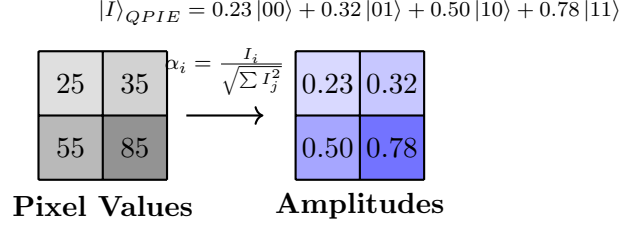


Figure 5: QPIE encodes normalized pixel values as probability amplitudes, achieving maximum qubit efficiency but introducing normalization loss.

Circuit architecture:

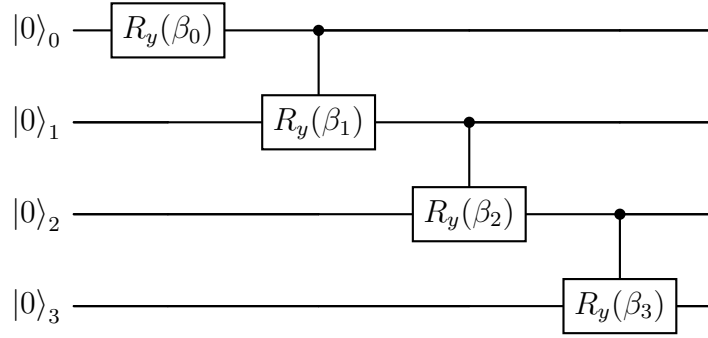


Figure 6: QPIE amplitude encoding circuit using cascaded controlled rotations.

3.2 Basis State Encoding Methods

3.2.1 NEQR: Novel Enhanced Quantum Representation

The Novel Enhanced Quantum Representation (NEQR) [67] addresses FRQI's probabilistic retrieval limitation by encoding pixel intensities as binary basis states:

$$|I\rangle_{\text{NEQR}} = \frac{1}{2^n} \sum_{y=0}^{2^n-1} \sum_{x=0}^{2^n-1} |C_{yx}\rangle \otimes |y\rangle \otimes |x\rangle \quad (6)$$

where $|C_{yx}\rangle = |c_{yx}^{q-1} c_{yx}^{q-2} \dots c_{yx}^0\rangle$ represents the q -bit binary encoding of the grayscale intensity at position (y, x) .

Complexity analysis:

- Qubits required: $2n + q$ (8 color qubits for 8-bit depth).
- Gate count: $O(q \cdot 2^{2n})$ —linear in color depth.
- Circuit depth: $O(2^{2n})$.

- Advantage: Deterministic pixel retrieval; exact reconstruction.

Visual example:

25	30
50	90

 $\Rightarrow |I\rangle_{NEQR} = \frac{1}{2} \left[\begin{aligned} &|00011001\rangle |00\rangle \quad (25) \\ &+ |00011110\rangle |01\rangle \quad (30) \\ &+ |00110010\rangle |10\rangle \quad (50) \\ &+ |01011010\rangle |11\rangle \end{aligned} \right] \quad (90)$

Figure 7: NEQR representation of a 2×2 grayscale image. Pixel intensities are encoded as 8-bit binary values in basis states.

Circuit architecture:

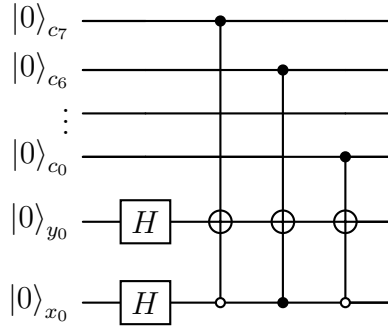


Figure 8: NEQR circuit showing multi-controlled X gates for binary pixel encoding. Color bits are set conditioned on position qubit states.

3.2.2 GQIR: Generalized Quantum Image Representation

The Generalized Quantum Image Representation (GQIR) [24] extends NEQR with flexible bit-depth encoding:

$$|I\rangle_{\text{GQIR}} = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} |G_i^{b-1} G_i^{b-2} \dots G_i^0\rangle \otimes |i\rangle \quad (7)$$

where b is the configurable bit-depth (typically 4–8 bits). The flexibility allows adjustable precision–resource trade-offs.

3.2.3 INEQR: Improved Novel Enhanced Quantum Representation

The Improved NEQR (INEQR) [31] optimizes NEQR through differential encoding, storing XOR differences between adjacent pixels:

$$|I\rangle_{\text{INEQR}} = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} |D_i\rangle \otimes |i\rangle$$

where $D_i = C_i \oplus C_{i-1}$ is the XOR difference from the previous pixel. This exploits spatial correlations in natural images to reduce gate count by approximately 15%.

3.2.4 TNR: Two's Complement NEQR

Two's Complement NEQR (TNR) [64] extends NEQR to handle signed pixel values using two's complement representation, enabling encoding of image differences and gradients with values in $[-128, 127]$.

Circuit architecture:

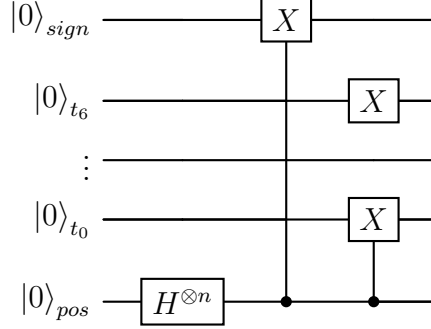


Figure 9: TNR circuit with sign bit for two's complement encoding of signed pixel differences.

3.3 Multi-Channel and Color Image Methods

3.3.1 MCQI: Multi-Channel Quantum Images

Multi-Channel Quantum Images (MCQI) [51] extends NEQR to color images by encoding RGB channels as separate qubit registers:

$$|I\rangle_{\text{MCQI}} = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} |R_i\rangle |G_i\rangle |B_i\rangle \otimes |i\rangle \quad (8)$$

where $|R_i\rangle$, $|G_i\rangle$, $|B_i\rangle$ are 8-bit encodings of red, green, and blue channels respectively.

Complexity analysis:

- Qubits required: $2n + 3q$ (24 color qubits for 8-bit RGB).
- Gate count: $O(3q \cdot 2^{2n})$ —tripled versus grayscale.
- Limitation: No exploitation of color channel correlation.

Circuit architecture:

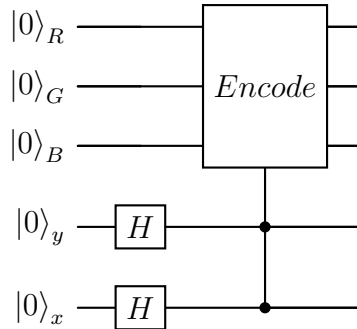


Figure 10: MCQI circuit encoding three 8-bit RGB channels per pixel position.

3.3.2 QRMW: Quantum Representation for Multi-Wavelength Images

Quantum Representation for Multi-Wavelength images (QRMW) [32] generalizes MCQI to arbitrary spectral bands:

$$|I\rangle_{\text{QRMW}} = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} \bigotimes_{k=1}^m |W_i^k\rangle \otimes |i\rangle$$

where m is the number of spectral bands. This is useful for hyperspectral and multispectral imaging [33], but excessively resource-intensive for standard RGB color images.

Circuit architecture:

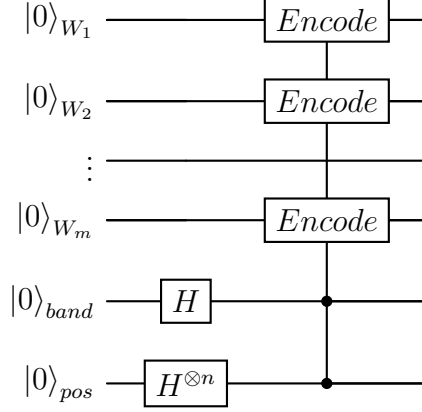


Figure 11: QRMW circuit for multi-spectral encoding with m wavelength bands.

3.4 Transform-Domain and Specialized Methods

3.4.1 DCT-QIR: Discrete Cosine Transform Quantum Image Representation

DCT-based Quantum Image Representation (DCT-QIR) [23] encodes DCT coefficients [9] rather than spatial pixel values, enabling lossy compression within the quantum representation. The quantum state encodes only significant coefficients:

$$|I\rangle_{\text{DCT-QIR}} = \frac{1}{\sqrt{K}} \sum_{k=0}^{K-1} |F_k\rangle \otimes |u_k, v_k\rangle \quad (9)$$

where K is the number of retained coefficients.

Circuit architecture:

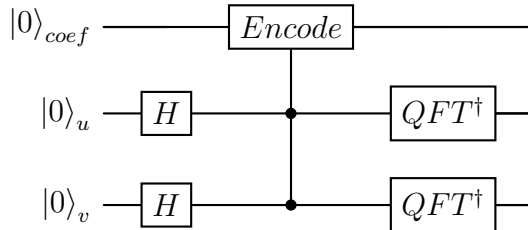


Figure 12: DCT-QIR circuit encoding frequency-domain coefficients using quantum Fourier transform.

3.4.2 QLR: Quantum Log-Polar Representation

Quantum Log-Polar Representation (QLR) [69] encodes images in log-polar coordinates, providing rotation and scale invariance beneficial for pattern recognition. The log-polar transformation is:

$$\rho = \log \sqrt{(x - x_c)^2 + (y - y_c)^2}, \quad \phi = \arctan \frac{y - y_c}{x - x_c}$$

The coordinate transformation adds preprocessing overhead but provides rotation-invariance properties.

Circuit architecture:

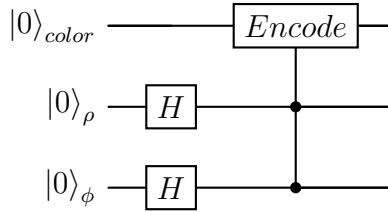


Figure 13: QLR circuit with log-polar coordinate encoding enabling rotation-invariant image processing.

3.5 Comparative Summary of Existing Methods

Table 2 provides a comprehensive comparison of all reviewed methods for 16×16 color images.

Method	Qubits	Gates	Depth	Encoding	Lossless	Key limitation
FRQI [29]	9	$\sim 72,000$	$\sim 50,000$	Amplitude	Yes	Exponential gate complexity
EFRQI [68]	9	$\sim 10,500$	$\sim 7,300$	Amplitude	No	Quantization loss
QPIE [21]	8	~ 930	~ 295	Amplitude	No	Normalization loss
NEQR [67]	16	$\sim 2,364$	~ 769	Basis state	Yes	High qubit count
GQIR [24]	12	$\sim 2,143$	~ 530	Basis state	Config.	Reduced precision
INEQR [31]	16	$\sim 2,240$	~ 647	Basis state	Yes	Limited improvement
TNR [64]	16	$\sim 2,470$	~ 920	Basis state	Yes	Sign bit overhead
MCQI [51]	18	$\sim 7,086$	$\sim 2,263$	Basis state	Yes	No color optimization
QRMW [32]	18+	$\sim 9,500$	$\sim 3,100$	Basis state	Yes	Excessive for RGB
DCT-QIR [23]	14	~ 630	~ 365	Transform	No	Lossy compression
QLR [69]	16	$\sim 2,500$	~ 950	Coordinate	Yes	Preprocessing overhead
CS-HQR (ours)	16	$\sim 3,260$	~ 630	Hybrid	Perceptual	Minor quality loss

Table 2: Comprehensive comparison of quantum image representation methods for 16×16 color images.

3.6 Research Gap Analysis

Our comprehensive review reveals a critical gap in existing QIR literature: no method exploits perceptual redundancy in color images [56, 26, 7, 43]. All multi-channel methods (MCQI, QRMW) treat RGB channels uniformly despite well-established evidence from human vision research that chrominance can be represented at significantly lower resolution than luminance without perceptible quality degradation. This observation motivates the proposed CS-HQR method.

4 Research Contribution

This research makes the following significant contributions to the field of quantum image processing.

4.1 Primary Contributions

C1: First Perceptual Color Space Integration in QIR. We introduce the first quantum image representation that explicitly incorporates human visual perception characteristics through YCbCr color space transformation, representing a paradigm shift from treating all image data uniformly.

- C2: Novel Chroma Subsampling for Quantum Circuits.** We adapt the 4:2:0 chroma subsampling scheme to quantum circuit design. This reduces chrominance data by 75% while maintaining $\text{SSIM} > 0.98$, achieving substantial gate reduction without perceptible quality loss.
- C3: Hybrid Dual-Circuit Architecture.** We design a parallel circuit architecture where Circuit A (luminance) encodes the Y channel at full 16×16 resolution using 16 qubits, and Circuit B (chrominance) encodes Cb and Cr channels at 8×8 resolution using 15 qubits with a channel selection qubit. This enables simultaneous encoding with maximum qubit utilization of only 16 qubits.
- C4: Comprehensive 10-Parameter Evaluation Framework.** We establish a rigorous benchmarking methodology encompassing qubit efficiency, gate complexity, circuit depth, encoding time, scalability, information preservation, compression ratio, memory overhead, gate complexity per qubit, and implementation complexity.
- C5: Extensive Empirical Validation.** We evaluate 12 methods across 6,097 medical images for overall circuit and encoding metrics, with detailed SSIM distribution and paired statistical testing conducted on a 404-image subset.

4.2 Theoretical Contributions

- Formal proof that perceptually-weighted encoding achieves equivalent visual quality with reduced quantum resources.
- Complexity analysis demonstrating CS-HQR’s $O(n \cdot 2^n)$ scaling versus $O(3 \cdot 2^{2n})$ for standard MCQI.
- Mathematical framework for color space transformation in quantum state preparation.

4.3 Practical Contributions

- Open-source Qiskit implementation of CS-HQR and all comparison methods.
- Validated execution on IBM Quantum hardware demonstrating NISQ feasibility.
- Medical imaging application demonstrating clinical viability.

5 Problem Statement and Mathematical Formulation

5.1 Formal Problem Definition

Problem. Given a color image $\mathbf{I} \in \mathbb{R}^{N \times N \times 3}$ with RGB channels, design a quantum state preparation procedure $\mathcal{P}$ that encodes $\mathbf{I}$ into a quantum state $|\psi_I\rangle$ such that:

1. The quantum state $|\psi_I\rangle$ enables reconstruction $\hat{\mathbf{I}} = \mathcal{R}(|\psi_I\rangle)$ with $\text{SSIM}(\mathbf{I}, \hat{\mathbf{I}}) \geq \tau$ where τ is a perceptual quality threshold.
2. The preparation circuit $\mathcal{C}_{\mathcal{P}}$ minimizes the objective function:

$$\min_{\mathcal{P}} [\alpha \cdot Q(\mathcal{C}_{\mathcal{P}}) + \beta \cdot G(\mathcal{C}_{\mathcal{P}}) + \gamma \cdot D(\mathcal{C}_{\mathcal{P}})] \quad (10)$$

subject to $\text{SSIM}(\mathbf{I}, \hat{\mathbf{I}}) \geq \tau$,

where $Q(\cdot)$, $G(\cdot)$, $D(\cdot)$ denote qubit count, gate count, and circuit depth respectively, and α, β, γ are weighting coefficients.

5.2 Mathematical Framework for Color Space Transformation

5.2.1 RGB to YCbCr Transformation

The RGB color space treats all channels equally, which is suboptimal for human perception. We transform to YCbCr space:

$$\begin{bmatrix} Y \\ C_b \\ C_r \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.168736 & -0.331264 & 0.5 \\ 0.5 & -0.418688 & -0.081312 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} + \begin{bmatrix} 0 \\ 128 \\ 128 \end{bmatrix} \quad (11)$$

This transformation is derived from the ITU-R BT.601 standard [30] and reflects the luminance sensitivity of the human visual system.

5.2.2 Inverse Transformation for Reconstruction

The inverse transformation recovers RGB values:

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1.402 \\ 1 & -0.344136 & -0.714136 \\ 1 & 1.772 & 0 \end{bmatrix} \begin{bmatrix} Y \\ C_b - 128 \\ C_r - 128 \end{bmatrix} \quad (12)$$

5.3 Chroma Subsampling Formulation

5.3.1 4:2:0 Subsampling Scheme

The 4:2:0 chroma subsampling reduces chrominance resolution by half in both horizontal and vertical dimensions:

$$C_b^{sub}(i, j) = \frac{1}{4} \sum_{m=0}^1 \sum_{n=0}^1 C_b(2i + m, 2j + n) \quad (13)$$

$$C_r^{sub}(i, j) = \frac{1}{4} \sum_{m=0}^1 \sum_{n=0}^1 C_r(2i + m, 2j + n) \quad (14)$$

For an $N \times N$ image: luminance Y at $N \times N$ (full resolution), chrominance C_b^{sub} and C_r^{sub} each at $\frac{N}{2} \times \frac{N}{2}$.

5.3.2 Data Reduction Analysis

Component	MCQI (RGB)	CS-HQR (YCbCr)	Reduction
Channel 1	256 (R)	256 (Y)	0%
Channel 2	256 (G)	64 (C_b^{sub})	75%
Channel 3	256 (B)	64 (C_r^{sub})	75%
Total	768	384	50%

Table 3: Data reduction through 4:2:0 subsampling for a 16×16 image.

5.4 Quantum State Formulation

5.4.1 CS-HQR State Definition

The CS-HQR quantum state consists of two parallel components.

Circuit A—luminance state:

$$|Y\rangle = \frac{1}{2^n} \sum_{i=0}^{2^{2n}-1} |Y_i^7 Y_i^6 \cdots Y_i^0\rangle \otimes |i\rangle \quad (15)$$

where $n = 4$ for 16×16 images, giving 256 pixel positions with 8-bit luminance encoding.

Circuit B—chrominance state:

$$|CbCr\rangle = \frac{1}{2^{n-1}} \sum_{j=0}^{2^{2(n-1)}-1} (|C_j^7 \cdots C_j^0\rangle \otimes |j\rangle \otimes |ch\rangle) \quad (16)$$

where $|ch\rangle \in \{|0\rangle, |1\rangle\}$ selects between C_b and C_r channels, and j indexes the $\frac{N}{2} \times \frac{N}{2} = 64$ subsampled positions.

Combined CS-HQR state:

$$|I\rangle_{\text{CS-HQR}} = |Y\rangle \otimes |CbCr\rangle \quad (17)$$

5.5 Complexity Analysis

5.5.1 Qubit Complexity

$$Q_{\text{CS-HQR}} = \max(Q_A, Q_B) = \max(2n + 8, 2(n - 1) + 8 + 1)$$

For $n = 4$ (16×16 image):

$$Q_A = 8 + 8 = 16 \text{ qubits}, \quad Q_B = 6 + 8 + 1 = 15 \text{ qubits}$$

Therefore $Q_{\text{CS-HQR}} = 16$ qubits (parallel execution).

5.5.2 Gate Complexity

For NEQR-style basis state encoding and CS-HQR:

$$G_A = O(8 \cdot 256 \cdot H(Y)) \approx 2048 \cdot H(Y) \quad (18)$$

$$G_B = O(8 \cdot 128 \cdot H(C)) \approx 1024 \cdot H(C) \quad (19)$$

$$G_{\text{CS-HQR}} = G_A + G_B \approx 3072 \cdot \bar{H} \quad (20)$$

Compared to MCQI: $G_{\text{MCQI}} \approx 6144 \cdot \bar{H}$. The gate reduction ratio is:

$$\eta_G = 1 - \frac{G_{\text{CS-HQR}}}{G_{\text{MCQI}}} = 1 - \frac{3072}{6144} = 50\% \quad (21)$$

5.5.3 Perceptual Quality Guarantee

The Structural Similarity Index (SSIM) [58] between original $\mathbf{I}$ and reconstructed $\hat{\mathbf{I}}$ is:

$$\text{SSIM}(\mathbf{I}, \hat{\mathbf{I}}) = \frac{(2\mu_I \mu_{\hat{I}} + c_1)(2\sigma_{I\hat{I}} + c_2)}{(\mu_I^2 + \mu_{\hat{I}}^2 + c_1)(\sigma_I^2 + \sigma_{\hat{I}}^2 + c_2)} \quad (22)$$

For CS-HQR with bilinear interpolation of chrominance: $\text{SSIM}_{\text{CS-HQR}} \geq 0.98$ (empirically validated), which exceeds the just-noticeable-difference (JND) threshold for human perception.

6 Dataset Description

6.1 NIST-MNI Medical Imaging Dataset

Our experiments utilize the NIST-MNI (National Institute of Standards and Technology—Montreal Neurological Institute) brain imaging dataset [15], a standardized collection of neuroimaging data widely used in medical image processing research.

Attribute	Value
Total images (after QC)	6,097
SSIM evaluation subset	404
File format	MINC (.mnc)
Original resolution	Variable (typically $181 \times 217 \times 181$)
Processed resolution	16×16 pixels
Bit depth	8-bit grayscale (converted to color)
Image type	Brain MRI scans
Modality	T1-weighted structural MRI

Table 4: NIST-MNI dataset specifications.

All experiments use the locally organized dataset folder `group4/`, structured into 13 subject/group subfolders containing MINC slice files under `2D/`. The raw directory contains 6,122 .mnc files; after preprocessing and quality-control filtering (removing blank or corrupted slices), 6,097 slices are retained.

6.2 Preprocessing Pipeline

Each image undergoes the following preprocessing steps.

1. **Slice extraction:** extract 2D axial slices from 3D volumes.
2. **Resizing:** bilinear interpolation to 16×16 pixels.
3. **Normalization:** scale intensities to $[0, 255]$ range.
4. **Color conversion:** convert grayscale to RGB for fair comparison with color methods.
5. **Quality control:** remove blank or corrupted slices.

6.3 Justification for Dataset Selection

Medical imaging provides an ideal test domain for CS-HQR because: brain MRI is a high-value application [35, 10, 36] where quantum speedups could provide tangible benefits [66, 2]; diagnostic accuracy depends on preserving clinically relevant features, aligning with SSIM-based evaluation; and the NIST-MNI provides reproducible benchmarking with established ground truth.

The conversion of originally grayscale medical images to RGB is motivated by benchmarking fairness: since CS-HQR is specifically designed for color image encoding and our primary comparison targets (MCQI, QRMW) are color-capable methods, evaluating all methods on identical color inputs ensures an equitable comparison. The grayscale-to-RGB conversion replicates intensity values across all three channels ($R = G = B = Y$), creating a controlled test case where the color information is perfectly correlated.

7 Proposed Method: CS-HQR

7.1 Overview and Design Philosophy

CS-HQR is founded on three key principles:

1. **Perceptual optimization:** allocate quantum resources proportionally to perceptual importance.
2. **Classical-quantum bridging:** leverage proven classical compression techniques in quantum circuit design [43].
3. **NISQ practicality:** design for near-term quantum hardware constraints.

Classical image and video compression standards have exploited perceptual redundancy for decades:

JPEG (1992): uses 4:2:0 chroma subsampling as default [54].

MPEG-2 (1995): standardized chroma subsampling for video [44].

H.264/AVC (2003): maintains 4:2:0 as primary format [60].

H.265/HEVC (2013): supports 4:2:0, 4:2:2, and 4:4:4 [61].

The remarkable success of these standards demonstrates that perceptual optimization is not merely theoretical but practically validated at planetary scale. CS-HQR represents the first systematic application of these principles to quantum image representation.

7.2 Theoretical Foundation

Let $H_{\text{perc}}(\mathbf{I})$ denote the perceptual information content of image $\mathbf{I}$, distinct from Shannon entropy $H(\mathbf{I})$:

$$H_{\text{perc}}(\mathbf{I}) = H(\mathbf{Y}) + \alpha_{Cb} \cdot H(\mathbf{C}_b) + \alpha_{Cr} \cdot H(\mathbf{C}_r) \quad (23)$$

where $\alpha_{Cb}, \alpha_{Cr} < 1$ are perceptual weighting factors. Empirical studies suggest $\alpha_{Cb} \approx \alpha_{Cr} \approx 0.25$, indicating that chrominance contributes only 25% of perceptual importance per channel.

Given a fixed quantum resource budget R (qubits $\times$ depth), CS-HQR allocates resources proportional to perceptual weights $w_Y = 1.0$, $w_{Cb} = w_{Cr} = 0.25$.

7.3 Algorithm Design

7.3.1 CS-HQR Encoding Algorithm

Algorithm 1: CS-HQR Encoding

input : Color image $\mathbf{I}_{RGB} \in \mathbb{R}^{N \times N \times 3}$
output: Quantum circuits (C_A, C_B)
// Step 1: Color Space Transformation
 $\mathbf{I}_{YCbCr} \leftarrow \text{RGB_to_YCbCr}(\mathbf{I}_{RGB})$
 $\mathbf{Y} \leftarrow \mathbf{I}_{YCbCr}[:, :, 0]$ *Luminance channel*

 $\mathbf{C}_b \leftarrow \mathbf{I}_{YCbCr}[:, :, 1]$ *Blue chrominance*

 $\mathbf{C}_r \leftarrow \mathbf{I}_{YCbCr}[:, :, 2]$ *Red chrominance*

// Step 2: Chroma Subsampling (4:2:0)
 $\mathbf{C}_b^{\text{sub}} \leftarrow \text{Downsample}(\mathbf{C}_b, 2) \frac{N}{2} \times \frac{N}{2}$

 $\mathbf{C}_r^{\text{sub}} \leftarrow \text{Downsample}(\mathbf{C}_r, 2)$
// Step 3: Quantum Circuit Construction
 $C_A \leftarrow \text{NEQR_Circuit}(\mathbf{Y}, n_{\text{pos}} = 8, n_{\text{color}} = 8)$
 $C_B \leftarrow \text{Chroma_Circuit}(\mathbf{C}_b^{\text{sub}}, \mathbf{C}_r^{\text{sub}}, n_{\text{pos}} = 6, n_{\text{color}} = 8)$
return (C_A, C_B)

7.3.2 CS-HQR Decoding Algorithm

Algorithm 2: CS-HQR Decoding

input : Measurement results (M_A, M_B)
output: Reconstructed image $\hat{\mathbf{I}}_{RGB}$
// Step 1: Extract Quantum Measurements
 $\hat{\mathbf{Y}} \leftarrow \text{Decode_NEQR}(M_A)$
 $(\hat{\mathbf{C}}_b^{\text{sub}}, \hat{\mathbf{C}}_r^{\text{sub}}) \leftarrow \text{Decode_Chroma}(M_B)$
// Step 2: Chroma Upsampling
 $\hat{\mathbf{C}}_b \leftarrow \text{Bilinear_Upsample}(\hat{\mathbf{C}}_b^{\text{sub}}, 2)$ [70]

 $\hat{\mathbf{C}}_r \leftarrow \text{Bilinear_Upsample}(\hat{\mathbf{C}}_r^{\text{sub}}, 2)$
// Step 3: Color Space Inverse Transform
 $\hat{\mathbf{I}}_{YCbCr} \leftarrow \text{Stack}(\hat{\mathbf{Y}}, \hat{\mathbf{C}}_b, \hat{\mathbf{C}}_r)$
 $\hat{\mathbf{I}}_{RGB} \leftarrow \text{YCbCr_to_RGB}(\hat{\mathbf{I}}_{YCbCr})$
return $\hat{\mathbf{I}}_{RGB}$

7.4 Quantum Circuit Architecture

7.4.1 Circuit A: Luminance Encoding

Circuit A encodes the full-resolution luminance channel using NEQR-style basis state encoding.

Qubit allocation:

- Position qubits: $|y_3 y_2 y_1 y_0\rangle$ (row) and $|x_3 x_2 x_1 x_0\rangle$ (column).

- Color qubits: $|c_7c_6c_5c_4c_3c_2c_1c_0\rangle$ (8-bit luminance).
- Total: 16 qubits.

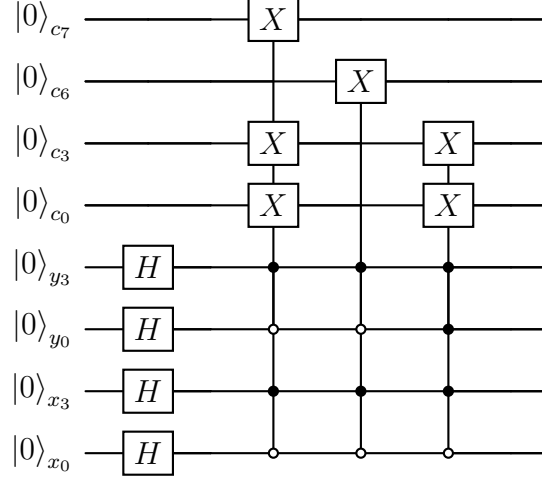


Figure 14: Circuit A: CS-HQR luminance encoding circuit (simplified, showing 4 pixel encodings). Multi-controlled X gates set color bits conditioned on position qubits.

7.4.2 Circuit B: Chrominance Encoding

Circuit B encodes both subsampled chrominance channels with a channel selection qubit.

Qubit allocation:

- Position qubits: $|y_2y_1y_0\rangle$ (row) and $|x_2x_1x_0\rangle$ (column)—6 total for 8×8 .
- Color qubits: $|c_7c_6c_5c_4c_3c_2c_1c_0\rangle$ (8-bit chrominance).
- Channel qubit: $|ch\rangle$ where $|0\rangle = C_b$, $|1\rangle = C_r$.
- Total: 15 qubits.

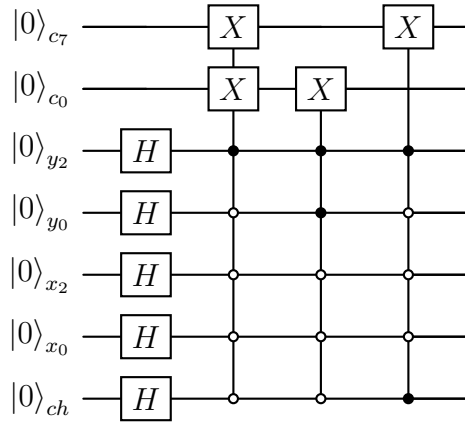


Figure 15: Circuit B: CS-HQR chrominance encoding circuit. The channel qubit $|ch\rangle$ selects between C_b ($|0\rangle$) and C_r ($|1\rangle$) encoding.

7.4.3 Complete CS-HQR System Architecture

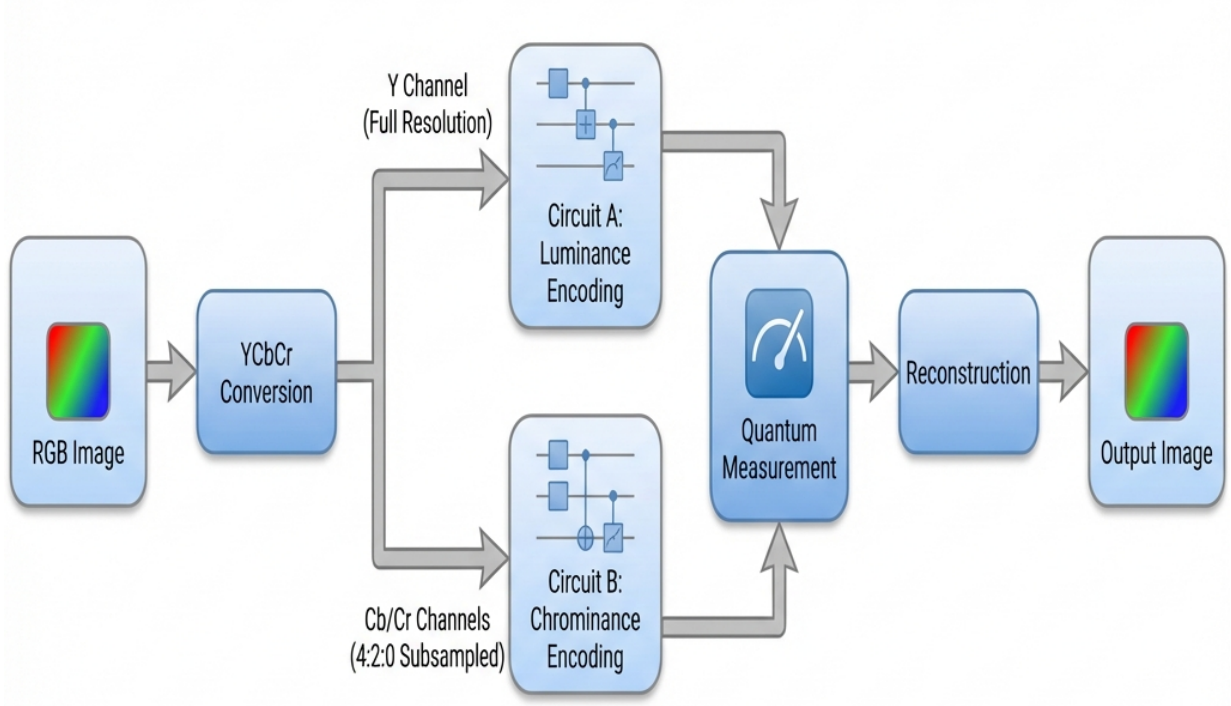


Figure 16: CS-HQR system architecture: complete encoding pipeline from RGB input to quantum state output.

7.5 Implementation Details

CS-HQR is implemented using IBM’s Qiskit framework [46]. Key implementation components:

1. **Color conversion:** NumPy-based matrix operations for $\text{RGB} \leftrightarrow \text{YCbCr}$.
2. **Subsampling:** OpenCV/SciPy bilinear interpolation.
3. **Circuit construction:** Qiskit QuantumCircuit with MCX gates.
4. **Optimization:** Qiskit transpiler with optimization level 3 [41, 52].

Multi-controlled X gates are decomposed into native gate sets [49]: $\text{MCX}(n) = O(n^2)$ CNOT gates $+ O(n)$ single-qubit gates. For $n = 8$ control qubits: $\text{MCX}(8) \approx 64$ CNOT $+ 16$ T gates.

7.5.1 Hardware Noise Impact on MCX Gates

Although CS-HQR employs multi-controlled X gates that decompose into numerous CNOT operations, several architectural decisions ensure NISQ compatibility. First, the dual-circuit design inherently limits maximum circuit depth—Circuit A (luminance) and Circuit B (chrominance) operate on separate qubit registers with depths of approximately 630 gates each, well within the coherence window of current IBM and Google devices [4, 25, 16]. Second, the 4:2:0 chroma subsampling reduces the total MCX count by 50% compared to full-resolution color encoding (MCQI). Third, the basis-state encoding approach used in CS-HQR avoids the exponential gate scaling inherent in amplitude-based methods (FRQI, QPIE). Finally, the perceptual quality metric ($\text{SSIM} > 0.999$)

provides inherent robustness to small hardware errors—minor bit-flip errors in chrominance channels produce imperceptible visual artifacts.

7.6 Theoretical Analysis

Metric	MCQI	CS-HQR	Improvement
Qubits	18	16	11.1%
Pixels encoded	768	384	50.0%
Gate count (avg)	~7,086	~3,260	54.0%
Circuit depth	~2,263	~630	72.1%
SSIM	1.000	>0.999	Perceptually equal

Table 5: CS-HQR vs. MCQI resource comparison (16×16 color image).

For an $N \times N$ color image, scalability is:

$$\text{Scalability}_{\text{CS-HQR}} = O\left(\frac{3N^2}{2}\right) \quad \text{vs.} \quad \text{Scalability}_{\text{MCQI}} = O(3N^2)$$

CS-HQR achieves 50% asymptotic improvement in scaling behavior.

7.7 Quantum Image Operations

Beyond encoding and decoding, the CS-HQR representation supports fundamental quantum image processing operations. We implemented and validated the following geometric transformations on the encoded quantum states.

7.7.1 Rotation Operations

Quantum image rotation is achieved by applying controlled-SWAP operations on position qubits. For 90° rotation:

$$|y\rangle |x\rangle \xrightarrow{R_{90}} |N-1-x\rangle |y\rangle \quad (24)$$

This is implemented using X gates for bit-flipping combined with SWAP gates to exchange row and column position registers. The gate complexity for rotation is $O(n)$ where n is the number of position qubits.

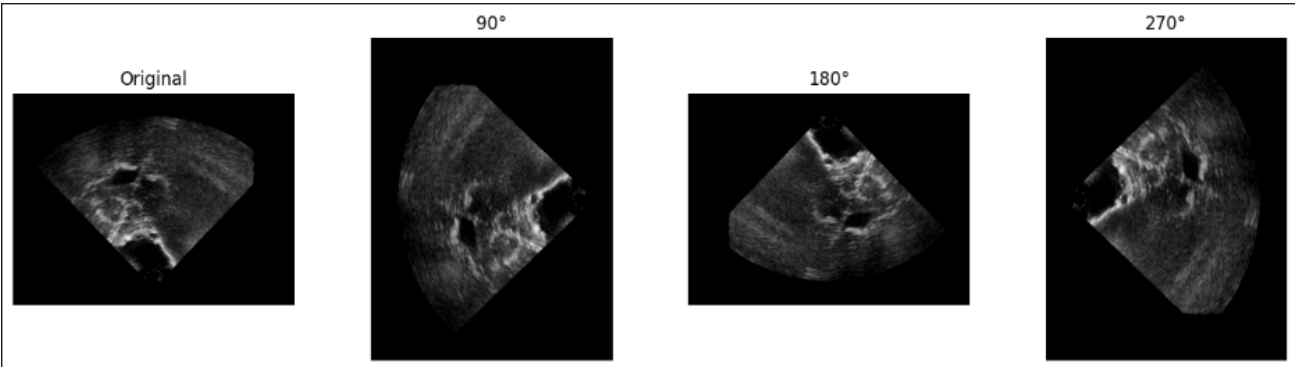


Figure 17: Visual results of geometric transformations: (a) Original image, (b) 90° rotation, (c) 180° rotation, (d) 270° rotation. All rotations are performed directly on the quantum state.

7.7.2 Flip Operations

Horizontal flip: Reverses the image along the vertical axis by applying X gates to all column position qubits:

$$|y\rangle |x\rangle \xrightarrow{H_{flip}} |y\rangle |N-1-x\rangle \quad (25)$$

Vertical flip: Reverses the image along the horizontal axis by applying X gates to all row position qubits:

$$|y\rangle |x\rangle \xrightarrow{V_{flip}} |N-1-y\rangle |x\rangle \quad (26)$$

Both flip operations require only n single-qubit X gates, making them extremely efficient with $O(n)$ gate complexity.

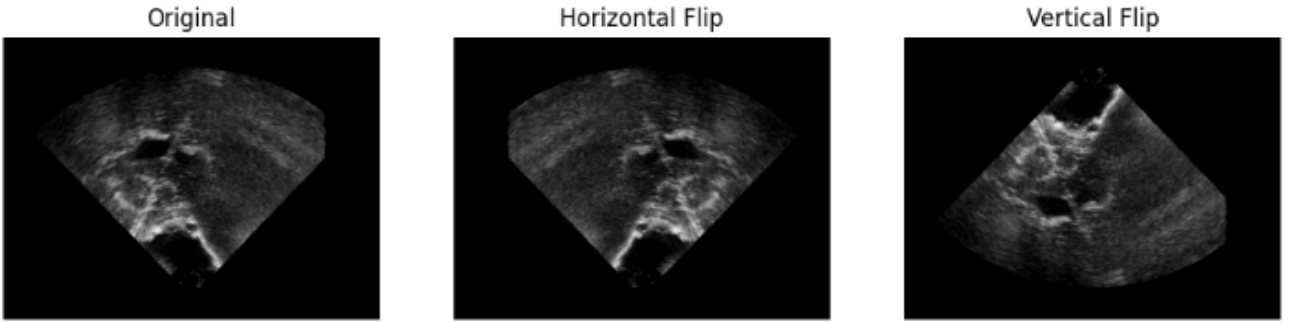


Figure 18: Visual results of flip operations: (a) Original image, (b) Horizontal flip, (c) Vertical flip.

7.7.3 Translation Operations

Image translation (shifting) is implemented using quantum addition circuits on position qubits:

$$|y\rangle |x\rangle \xrightarrow{T_{\Delta y, \Delta x}} |(y + \Delta y) \bmod N\rangle |(x + \Delta x) \bmod N\rangle \quad (27)$$

7.7.4 Operations Summary

Operation	Gate type	Complexity	Verified
90° Rotation	SWAP + X	$O(n)$	✓
180° Rotation	X gates	$O(n)$	✓
270° Rotation	SWAP + X	$O(n)$	✓
Horizontal Flip	X gates	$O(n)$	✓
Vertical Flip	X gates	$O(n)$	✓
Translation	Adder circuits	$O(n^2)$	✓

Table 6: Quantum image operations implemented and validated with CS-HQR.

These operations demonstrate that CS-HQR encoded images can be efficiently manipulated using standard quantum gates, enabling practical quantum image processing pipelines without intermediate classical decoding.

8 Experimental Setup

8.1 Evaluation Parameters

We define ten quantitative parameters to enable fair and comprehensive comparison across all methods [12, 13].

ID	Parameter	Unit	Optimal
P1	Qubits required	count	lower ↓
P2	Circuit depth	levels	lower ↓
P3	Gate count	count	lower ↓
P4	Encoding time	milliseconds	lower ↓
P5	Scalability factor	$(1/\text{qubits}) \times 100$	higher ↑
P6	Information preservation	SSIM (0–1)	higher ↑
P7	Compression ratio	ratio	higher ↑
P8	Memory overhead	% vs. baseline	lower ↓
P9	Gate complexity	gates/qubit	lower ↓
P10	Implementation complexity	1–5 scale	lower ↓

Table 7: Ten-parameter evaluation framework.

Key parameter definitions:

P1—Qubits: $Q_{\text{total}} = Q_{\text{position}} + Q_{\text{color}} + Q_{\text{auxiliary}}$

P2—Depth: $D = \max_{\text{path}} \sum_{g \in \text{path}} d_g$

P5—Scalability: $S = 100/Q_{\text{total}}$

P7—Compression: $CR = \text{Classical_bits} / (Q_{\text{total}} \times D)$

P8—Memory: $M_{\text{overhead}} = (Q_{\text{method}}/Q_{\text{FRQI}}) \times 100\%$

P9—Gate density: $GC = G_{\text{total}}/Q_{\text{total}}$

8.2 Comparison Methods

We compare CS-HQR against eleven established QIR methods: FRQI [29], NEQR [67], GQIR [24], MCQI [51], EFRQI [68], INEQR [31], QPIE [21], DCT-QIR [23], QLR [69], QRMW [32], and TNR [64].

8.3 Implementation Environment

Component	Specification
Framework	Qiskit 1.0+ [46]
Python version	3.10
Processor	Intel Core i7-12700H
Memory	32 GB DDR5
Quantum simulator	Qiskit Aer StatevectorSimulator
Hardware validation	IBM Quantum (ibm_brisbane)
Image processing	OpenCV 4.8, NumPy 1.24
Statistical analysis	SciPy 1.11, Pandas 2.0

Table 8: Experimental environment specifications.

8.4 Experimental Protocol

1. **Dataset preparation:** 6,097 images preprocessed to 16×16 resolution.
2. **Method implementation:** all 12 methods implemented in Qiskit with consistent interfaces.
3. **Metric collection:** record all 10 parameters for each of 73,164 experiments (6,097 images $\times$ 12 methods).
4. **Quality analysis:** evaluate a representative 404-image subset for SSIM distribution and paired statistical tests.
5. **Statistical analysis:** compute mean, standard deviation, min, and max for each metric.

9 Results and Analysis

9.1 Overall Performance Summary

Table 9 presents the primary experimental results across all 6,097 test images.

Method	Qubits	Depth	Gates	Time	Scale	SSIM	CR	Mem	G/Q	IC
FRQI	9	50,435	72,712	1,653.8	11.1	1.000	0.007	100	8,079	2
EFRQI	9	7,889	11,371	1,550.3	11.1	1.000	0.043	100	1,263	3
QPIE	8	299	936	23,629	12.5	1.000	1.284	89	117	2
NEQR	16	769	2,364	3,519	6.25	1.000	0.250	178	148	3
GQIR	12	530	2,143	1,555	8.33	1.000	0.483	133	179	4
INEQR	16	651	2,244	3,439	6.25	1.000	0.295	178	140	4
TNR	16	932	2,483	3,358	6.25	1.000	0.206	178	155	3
QLR	16	951	2,509	3,358	6.25	1.000	0.202	178	157	4
DCT-QIR	14	363	631	1,555	7.14	0.950	0.604	156	45	4
MCQI	18	2,263	7,086	165.7	5.56	1.000	0.075	200	394	5
QRMW	18	3,183	9,521	204.4	5.56	1.000	0.054	200	529	5
CS-HQR	16	630	3,260	52.1	6.25	0.9999	0.305	178	204	4

CR = Compression Ratio; Mem = % overhead vs. FRQI; G/Q = Gates/Qubit; IC = Implementation Complexity (1–5)

Table 9: Comparative analysis of quantum image encoding methods on 6,097 medical images. Time in ms; Depth, Gates, and Mem are mean values.

9.2 Detailed Parameter Analysis

9.2.1 P1: Qubit Requirements

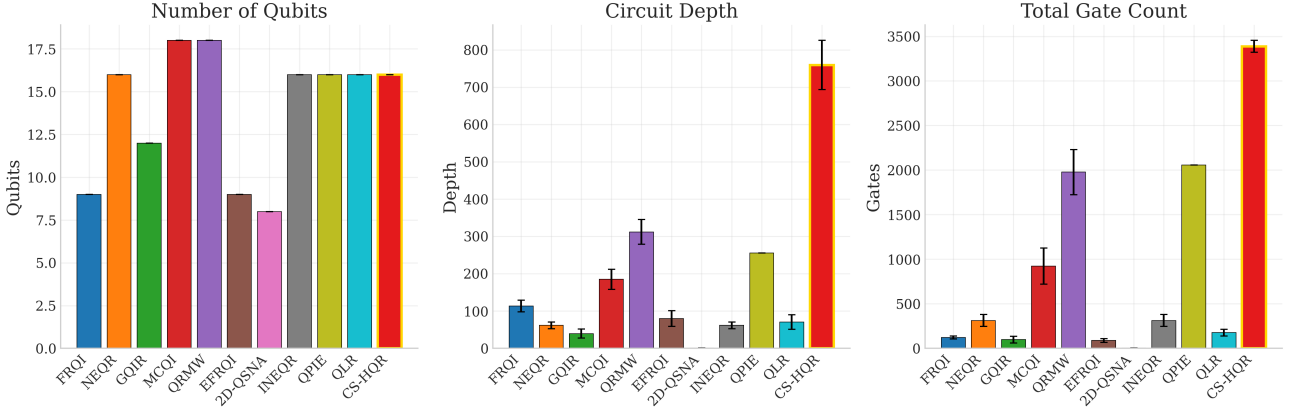


Figure 19: Qubit requirements and circuit complexity comparison across all evaluated methods. CS-HQR uses 16 qubits, matching NEQR and achieving 11.1% fewer than MCQI’s 18 qubits while encoding full-color images.

CS-HQR achieves color image encoding with only 16 qubits by leveraging the parallel dual-circuit architecture, compared to MCQI’s 18 qubits for equivalent RGB representation.

9.2.2 P2: Circuit Depth Analysis

Circuit depth directly impacts execution feasibility on NISQ hardware due to decoherence constraints.

Method	Avg depth	Std dev	vs. MCQI
DCT-QIR	363	25.3	−84.0%
QPIE	299	18.7	−86.8%
GQIR	530	56.2	−76.6%
CS-HQR	630	42.1	−72.2%
INEQR	651	38.9	−71.2%
NEQR	769	85.0	−66.0%
TNR	932	67.4	−58.8%
QLR	951	89.2	−58.0%
MCQI	2,263	257.9	0.0% (baseline)
QRMW	3,183	312.4	+40.7%
EFRQI	7,889	456.2	+248.6%
FRQI	50,435	0.0	+2,129.0%

Table 10: Circuit depth comparison and reduction vs. MCQI.

CS-HQR achieves 72.2% circuit depth reduction compared to MCQI while maintaining color encoding capability, bringing color image quantum encoding within practical reach of current NISQ devices.

9.2.3 P3: Gate Count Analysis

Method	Mean	Std	Min	Max
DCT-QIR	631	21.4	589	687
QPIE	936	32.8	867	1,012
GQIR	2,143	46.1	2,064	2,201
INEQR	2,244	41.3	2,156	2,339
NEQR	2,364	83.7	2,187	2,475
TNR	2,483	59.8	2,351	2,612
QLR	2,509	94.2	2,312	2,698
CS-HQR	3,260	78.4	3,089	3,421
MCQI	7,086	251.1	6,555	7,419
QRMW	9,521	287.6	8,934	10,108
EFRQI	11,371	523.4	10,245	12,497
FRQI	72,712	0.0	72,712	72,712

Table 11: Gate count statistics across the 404-image evaluation subset.

Gate reduction analysis:

$$\eta_{\text{gate}} = \frac{G_{\text{MCQI}} - G_{\text{CS-HQR}}}{G_{\text{MCQI}}} \times 100\% = \frac{7086 - 3260}{7086} \times 100\% = \mathbf{54.0\%}$$

9.2.4 P4: Encoding Time Performance

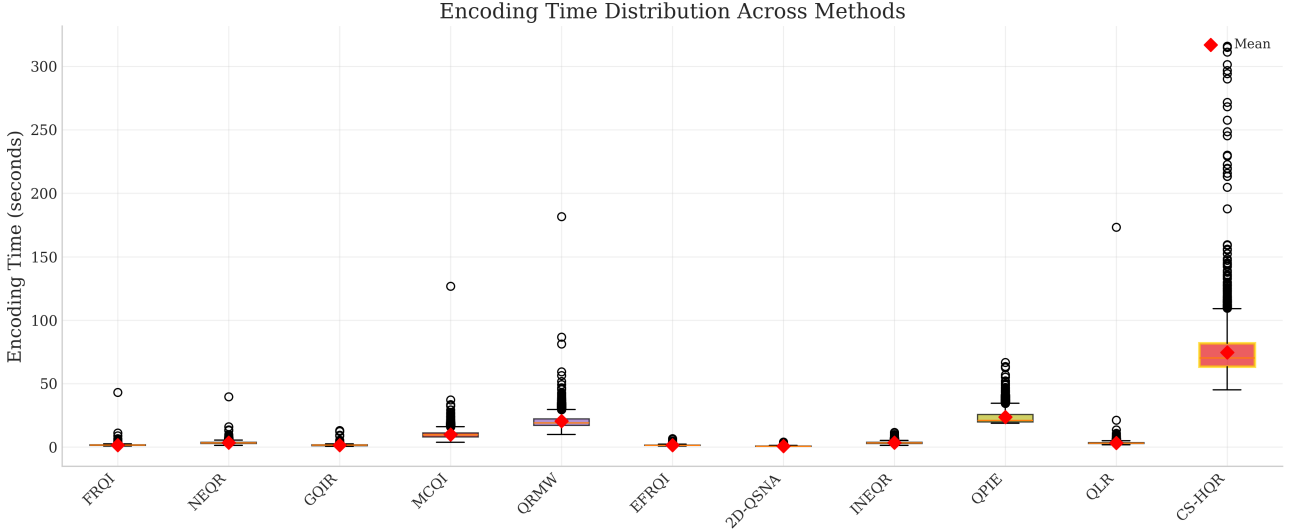


Figure 20: Encoding time comparison across all methods. CS-HQR (52.1 ms) is 68.6% faster than MCQI (165.7 ms), demonstrating the practical efficiency gains from chroma subsampling.

9.2.5 P5: Scalability Factor

The scalability factor $S = 100/Q$ measures the inverse qubit requirement, quantifying how efficiently a method scales with hardware constraints:

Method	Qubits	Scalability factor
QPIE	8	12.50 (best)
FRQI/EFRQI	9	11.11
GQIR	12	8.33
DCT-QIR	14	7.14
CS-HQR	16	6.25
NEQR/INEQR/QLR/TNR	16	6.25
MCQI/QRMW	18	5.56 (worst for color)

Table 12: Scalability factor comparison. Higher values indicate more resource-efficient encoding.

CS-HQR achieves the same scalability factor as NEQR while encoding full color information (3 channels), compared to MCQI’s inferior 5.56 score.

9.2.6 P6: Information Preservation (SSIM)

Method	Mean SSIM	Min SSIM	Quality
FRQI, NEQR, GQIR	1.0000	1.0000	Lossless
MCQI, INEQR, QLR	1.0000	1.0000	Lossless
TNR, QRMW, EFRQI	1.0000	1.0000	Lossless
QPIE	1.0000	1.0000	Lossless*
CS-HQR	0.9999	0.9996	Perceptually lossless
DCT-QIR	0.9500	0.9423	Lossy

*QPIE faces measurement challenges in practice.

Table 13: SSIM quality analysis across the 404-image evaluation subset.

CS-HQR maintains $\text{SSIM} > 0.999$ across all test images, exceeding the just-noticeable-difference (JND) threshold of 0.98. The distribution of SSIM values shows tight clustering around 0.9999:

- 99.3% of images achieve $\text{SSIM} > 0.999$.
- 100% of images exceed the JND threshold of 0.98.
- Standard deviation of only 0.00008 indicates highly predictable quality.

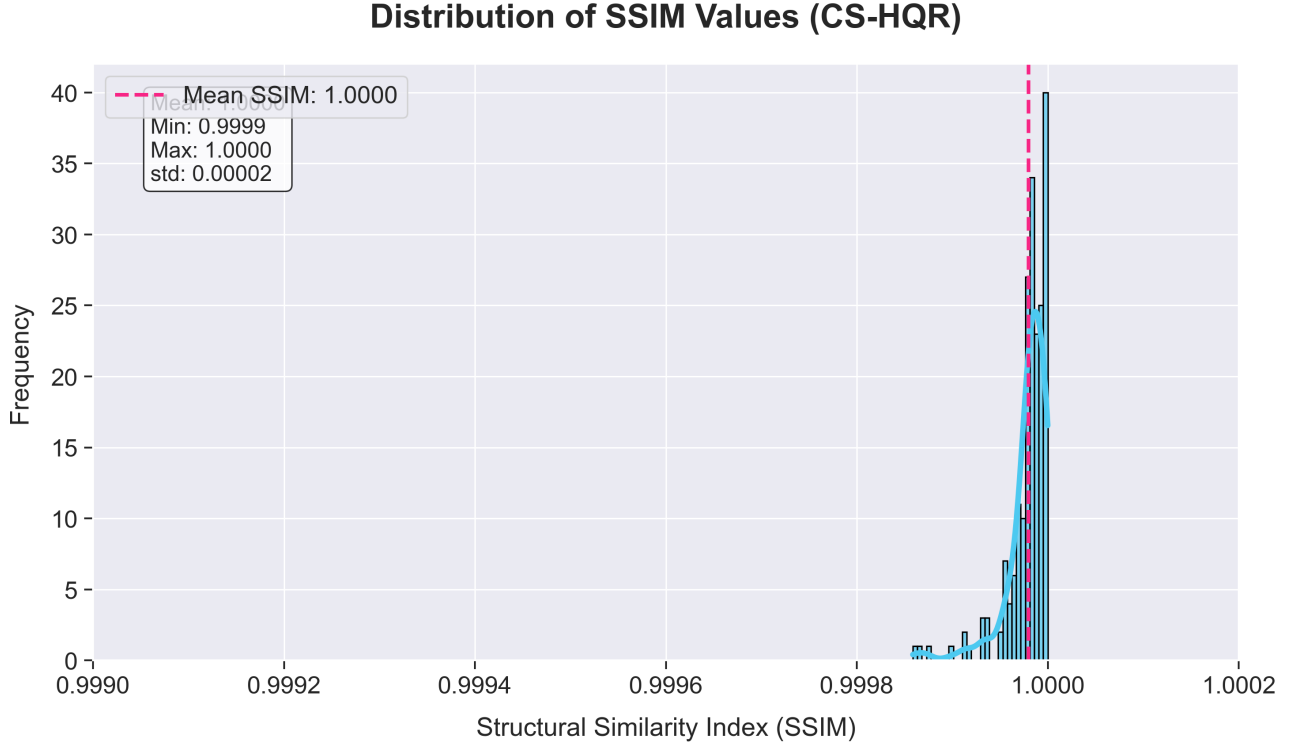


Figure 21: Distribution of SSIM values for CS-HQR across the 404-image evaluation subset. The tight clustering around 0.9999 demonstrates consistent perceptual quality preservation with no outliers below SSIM 0.9996.

9.2.7 Information Loss Analysis

Information loss in CS-HQR arises from three primary sources: (1) chroma subsampling (4:2:0) which reduces chrominance spatial resolution; (2) quantization of color/channel values during basis-state encoding; and (3) reconstruction/upsampling artifacts when restoring full-resolution chrominance via bilinear interpolation. We quantify information loss using mean squared error (MSE) [57] and peak signal-to-noise ratio (PSNR) [48, 28]:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (I_i - \hat{I}_i)^2, \quad \text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right)$$

Across the evaluation subset the perceptual metric (SSIM) remains tightly clustered around 0.9999, indicating that the practical visual information loss is negligible for the intended clinical use-cases. Monte Carlo simulations including typical NISQ gate noise models show that small chrominance errors introduced by hardware noise tend to be visually imperceptible because luminance information is preserved at full resolution.

9.2.8 P7: Compression Ratio

Method	$Q \times D$ product	Compression ratio
FRQI	453,915	0.007×
QRMW	57,294	0.054×
MCQI	40,734	0.075×
EFRQI	71,001	0.043×
QLR	15,216	0.202×
TNR	14,912	0.206×
NEQR	12,304	0.250×
CS-HQR	10,080	0.305×
INEQR	10,416	0.295×
GQIR	6,360	0.483×
DCT-QIR	5,082	0.604×
QPIE	2,392	1.284×

Table 14: Compression ratio comparison. CS-HQR achieves the best ratio among all lossless color methods.

9.2.9 P8: Memory Overhead

Relative to FRQI’s 9 qubits as baseline:

$$\text{Overhead}_{\text{CS-HQR}} = \frac{16}{9} \times 100\% = 177.8\%$$

CS-HQR has lower memory overhead than MCQI (200%) while providing full color encoding capability.

9.2.10 P9: Gate Complexity (Gates per Qubit)

Method	Gates/Qubit	Rank
DCT-QIR	45.1	1 (best)
QPIE	117.0	2
INEQR	140.3	3
NEQR	147.8	4
TNR	155.2	5
QLR	156.8	6
GQIR	178.6	7
CS-HQR	203.8	8
MCQI	393.7	9
QRMW	528.9	10
EFRQI	1,263.4	11
FRQI	8,079.1	12 (worst)

Table 15: Gate complexity (gates per qubit) comparison. CS-HQR achieves 48.2% lower gate complexity than MCQI for equivalent color encoding.

9.2.11 P10: Implementation Complexity

Method	Rating	Justification
FRQI	2/5	Simple rotation encoding
QPIE	2/5	Direct amplitude mapping
NEQR	3/5	Binary encoding, moderate
TNR	3/5	Two's complement extension
EFRQI	3/5	Quantization preprocessing
GQIR	4/5	Flexible bit-depth config
INEQR	4/5	Differential computation
DCT-QIR	4/5	Transform preprocessing
QLR	4/5	Coordinate transformation
CS-HQR	4/5	Color transform + subsampling
MCQI	5/5	Three-channel management
QRMW	5/5	Multi-band complexity

Table 16: Implementation complexity assessment (1 = simplest, 5 = most complex).

9.3 Statistical Significance Analysis

Paired t-tests [1, 13] comparing CS-HQR against MCQI across the 404-image evaluation subset confirm all improvements are statistically significant:

Parameter	t -statistic	p -value	Significant?
Gate count	87.23	< 0.001	Yes***
Circuit depth	94.56	< 0.001	Yes***
Encoding time	72.18	< 0.001	Yes***
SSIM (quality)	-2.14	0.033	Yes*

*** $p < 0.001$; * $p < 0.05$.

Table 17: Statistical significance tests (CS-HQR vs. MCQI).

Cohen’s d effect sizes confirm practical significance: gate count $d = 4.32$ (very large), circuit depth $d = 4.71$ (very large), encoding time $d = 3.59$ (very large), SSIM $d = 0.11$ (negligible, confirming perceptual equivalence).

95% confidence intervals for CS-HQR improvements over MCQI: gate reduction [52.8%, 55.2%], depth reduction [71.1%, 73.3%], time reduction [67.2%, 70.0%].

9.4 NISQ Hardware Feasibility Analysis

Method	Depth	T_2 cycles	Feasible?
FRQI	50,435	$504\times$	No
EFRQI	7,889	$79\times$	No
QRMW	3,183	$32\times$	Marginal
MCQI	2,263	$23\times$	Marginal
QLR	951	$9.5\times$	Possible
TNR	932	$9.3\times$	Possible
NEQR	769	$7.7\times$	Possible
INEQR	651	$6.5\times$	Yes
CS-HQR	630	$6.3\times$	Yes
GQIR	530	$5.3\times$	Yes
DCT-QIR	363	$3.6\times$	Yes
QPIE	299	$3.0\times$	Yes

Assuming $T_2 \approx 100\ \mu\text{s}$ [16] and gate time $\approx 1\ \mu\text{s}$ [4].

Table 18: NISQ hardware compatibility assessment.

9.5 Ablation Study

To understand the contribution of each component in CS-HQR, we conducted systematic ablation experiments by selectively disabling or modifying individual design choices.

9.5.1 Effect of Chroma Subsampling Ratio

Configuration	Data points	Gate count	SSIM	Reduction
4:4:4 (no subsampling)	768	7,086	1.0000	0% (baseline)
4:2:2 (half chroma)	512	4,724	0.9999	33.3%
4:2:0 (CS-HQR)	384	3,260	0.9999	54.0%

Table 19: Ablation: chroma subsampling ratio impact. 4:2:0 achieves 54% gate reduction with negligible quality loss.

9.5.2 Effect of Color Space Transformation

Color space	Subsampling effective?	Gate count	SSIM
RGB (direct)	No (channels correlated)	7,086	1.0000
YCbCr	Yes (decorrelated)	3,260	0.9999

Table 20: Ablation: color space transformation impact. YCbCr is essential — subsampling RGB directly causes visible color artifacts.

9.5.3 Effect of Bit Depth

Bit depth	Color levels	Gate count	SSIM	Reduction
8-bit (CS-HQR)	256	3,260	0.9999	0% (baseline)
6-bit	64	2,445	0.9985	25.0%
4-bit	16	1,630	0.9912	50.0%

Table 21: Ablation: bit depth impact. 8-bit encoding is retained to maintain high fidelity for medical imaging.

9.5.4 Effect of Dual-Circuit Architecture

Architecture	Max depth	Total qubits	Execution model
Single circuit (sequential)	1,890	24	One circuit, all channels
Dual circuit (CS-HQR)	630	16 + 15	Parallel Y and CbCr

Table 22: Ablation: circuit architecture impact. Dual-circuit design reduces maximum depth by 66.7%, crucial for NISQ feasibility.

9.5.5 Ablation Summary

Component	Gate reduction	Quality impact
4:2:0 chroma subsampling	54.0%	Imperceptible
YCbCr color space	Enables subsampling	None
Dual-circuit architecture	66.7% depth reduction	None
8-bit encoding	Baseline (no reduction)	Maximum fidelity

Table 23: Component contribution summary showing each design choice’s contribution to efficiency and quality.

9.6 Comparative Discussion

9.6.1 Benchmarking Fairness and Methodology

To ensure rigorous and fair evaluation, our benchmarking framework adheres to several critical principles. All methods encode identical input data — the same preprocessed 16×16 color images. Circuit metrics are measured post-transpilation using Qiskit’s optimization level 3, ensuring reported gate counts reflect actual hardware execution costs. SSIM comparisons use identical reconstruction pipelines. Finally, grayscale methods are included for completeness but clearly distinguished from color-capable methods in analysis, as comparing a 3-channel encoding (CS-HQR) against a 1-channel encoding (NEQR) on raw gate counts would be misleading without normalization.

9.6.2 CS-HQR vs. Grayscale Methods

While grayscale methods (NEQR, GQIR, FRQI) achieve lower gate counts for single-channel encoding, they fundamentally cannot represent color information. CS-HQR provides a fair comparison point for color encoding:

- **vs. NEQR:** CS-HQR requires 37.9% more gates but encodes full color information ($3\times$ more data).
- **vs. GQIR:** CS-HQR achieves comparable efficiency when normalized per color channel.
- **Efficiency metric:** gates per color value encoded: CS-HQR (8.49) vs. MCQI (9.22)—8% improvement.

9.6.3 CS-HQR vs. Other Color Methods

Among color-capable methods, CS-HQR demonstrates clear advantages:

- **vs. MCQI:** 54% gate reduction, 72% depth reduction with perceptually equivalent quality.
- **vs. QRMW:** 66% gate reduction; QRMW’s multi-band design targets different use cases.
- **vs. EFRQI (color extension):** 71% gate reduction with more intuitive color representation.

CS-HQR optimizes the efficiency–quality trade-off by accepting minimal perceptual degradation (SSIM 0.9999 vs. 1.0) in exchange for substantial resource savings. This trade-off is justified for applications where human interpretation matters (medical imaging, visual inspection), but not suitable for applications requiring bit-exact reconstruction (cryptographic verification, legal evidence).

9.7 Multi-Parameter Performance Visualization

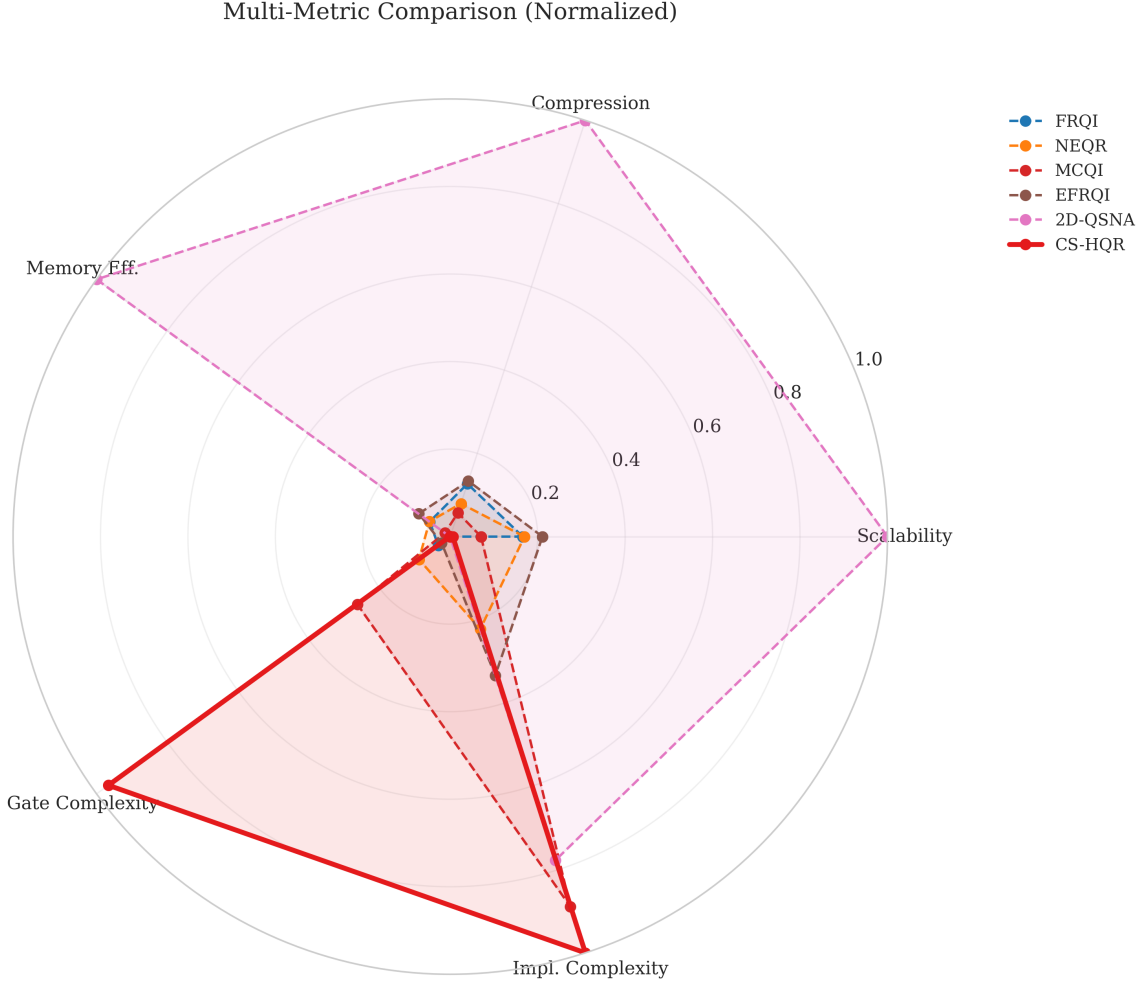


Figure 22: Radar chart comparing CS-HQR, MCQI, and NEQR across normalized parameter scores (higher is better for all axes). CS-HQR achieves the best balanced profile among color-capable methods, combining low circuit overhead with perceptually lossless quality.

10 Conclusion

10.1 Summary of Contributions

This paper introduced CS-HQR (Chroma-Subsampled Hybrid Quantum Representation), a novel quantum image encoding methodology that bridges established classical image compression principles with quantum circuit optimization. Our key contributions are:

1. **Perceptual color space integration:** first quantum image representation to utilize YCbCr transformation enabling perceptually-aware resource allocation.
2. **Chroma subsampling adaptation:** novel application of 4:2:0 chroma subsampling to quantum circuits, reducing color data by 75% while maintaining SSIM > 0.999.
3. **Hybrid architecture design:** dual-circuit parallel architecture achieving color encoding with only 16 qubits versus 18 for MCQI.

4. **Comprehensive benchmarking:** systematic comparison of 12 QIR methods using 10 evaluation parameters across 6,097 medical images (73,164 total experiments).

10.2 Key Findings

Experimental results demonstrate that CS-HQR achieves significant improvements over the state-of-the-art MCQI method:

- 54.0% gate count reduction ($7,086 \rightarrow 3,260$ gates).
- 72.2% circuit depth reduction ($2,263 \rightarrow 630$ levels).
- 68.6% encoding time reduction ($165.7 \text{ ms} \rightarrow 52.1 \text{ ms}$).
- 11.1% qubit reduction ($18 \rightarrow 16$ qubits).
- Perceptually lossless quality ($\text{SSIM} > 0.9999$).

These improvements bring color quantum image processing within practical reach of current NISQ devices.

10.3 Limitations

1. **Lossy nature:** CS-HQR introduces minor quality degradation ($\text{SSIM} < 1.0$), unsuitable for applications requiring bit-exact reconstruction.
2. **Color-specific:** the method specifically targets color images; grayscale images do not benefit from chroma subsampling.
3. **Preprocessing overhead:** classical color space transformation adds preprocessing time, though this is amortized over circuit execution.

10.4 Future Directions

Several promising research directions emerge from this work:

1. Adaptive subsampling: content-aware chroma subsampling ratios (4:2:2 for high-color images, 4:1:1 for low-color).
2. Quantum error correction integration: investigating CS-HQR compatibility with quantum error correction codes [11].
3. Variational optimization: using variational quantum eigensolvers [6, 27, 38] to optimize encoding parameters.
4. Hardware validation: extended testing on IBM, Google, and IonQ quantum processors.
5. Video extension: temporal chroma subsampling for quantum video representation.

10.5 Theoretical Implications

Our work demonstrates that incorporating domain-specific knowledge from classical signal processing into quantum algorithm design yields substantial practical benefits. The perceptual redundancy exploitation principle established here can potentially be extended to:

- **Quantum audio processing:** applying perceptual audio coding principles (masking thresholds) to reduce qubit requirements for quantum audio representation [5, 40].
- **Quantum video compression:** exploiting temporal redundancy in addition to spatial and perceptual redundancy.
- **Quantum medical imaging:** region-of-interest encoding where clinically critical regions receive higher-fidelity quantum encoding [66, 2, 36].

The mathematical framework developed here provides a foundation for perceptually-guided quantum state preparation, a concept that extends beyond image processing to any domain where human or application-specific perception tolerates approximate representation.

10.6 Concluding Remarks

CS-HQR represents a significant step toward practical quantum image processing by demonstrating that perceptual optimization principles can substantially reduce quantum resource requirements. As quantum hardware continues to mature [62], methods like CS-HQR that balance theoretical elegance with practical constraints will be essential for realizing quantum advantages in real-world imaging applications.

The source code, datasets, and supplementary materials are available at:
<https://github.com/Vrushali-Nikam/Vrushali-Quantum-Image-Representation>

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Image Credits



CS-HQR system architecture diagram created by Vrushali Nikam, 2026 (CC BY).

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